higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING

(First Paper)

NQF LEVEL 4

(12050674)

13 November 2019 (X-Paper)
09:00–11:00

This question paper consists of 6 pages, 1 addendum and 1 formula sheet.

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TIME: 2 HOURS
MARKS: 75

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Use only a BLACK or BLUE ink.
6. Write neatly and legibly.

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QUESTION 1: REINFORCED-CONCRETE DETAILING AND SPECIFICATIONS

1.1 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–L) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.1.1	Tendency of a force to cause rotation about a point	A	newton
1.1.2	Unit of stress	B	shear force
1.1.3	Equal to 1 kPa	C	moment of force
1.1.4	Distribution of forces that works over the entire area	D	compressive force
1.1.5	Unit of force	E	axial force
1.1.6	Works perpendicular to the axis under consideration	F	N/m^2
1.1.7	Same as gravitational force	G	weight
1.1.8	Another name for vector addition	H	pascal
1.1.9	Pushing force	I	$1\ kN/m^2$
1.1.10	Works parallel to the axis under consideration	J	triangle of forces
		K	stress
		L	tensile force

(10 × 1) (10)

1.2 Calculate the resultant force and direction shown in FIGURE 1.

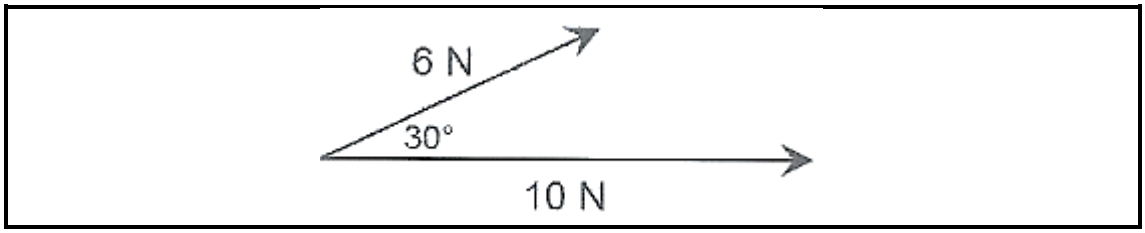


FIGURE 1 (3)

1.3 Explain the following description for dimensioning reinforcing bars:

16 R12 – 05 -150T (6 × ½) (3)

1.4 Differentiate between the TWO loads illustrated in FIGURE 2.

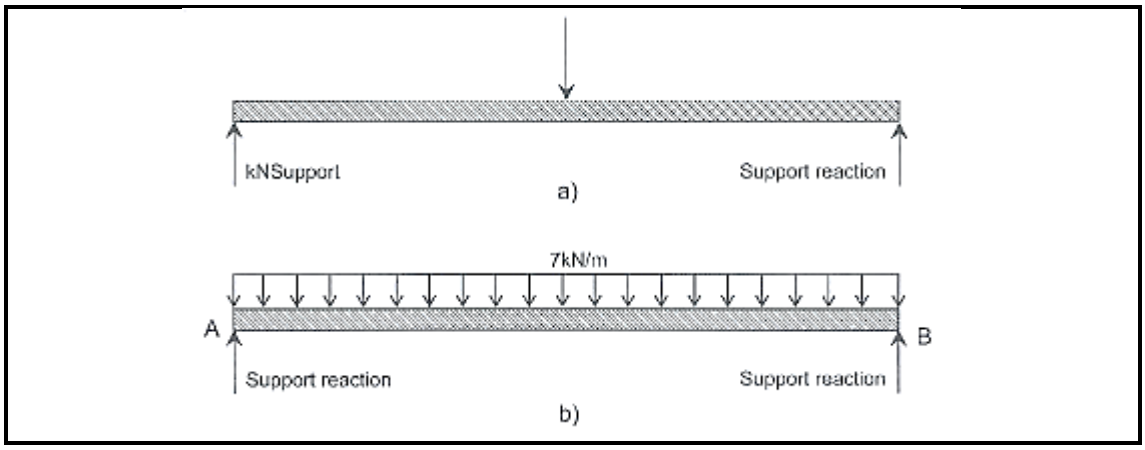


FIGURE 2 (2)

1.5 Draw a shear-force diagram from FIGURE 3.

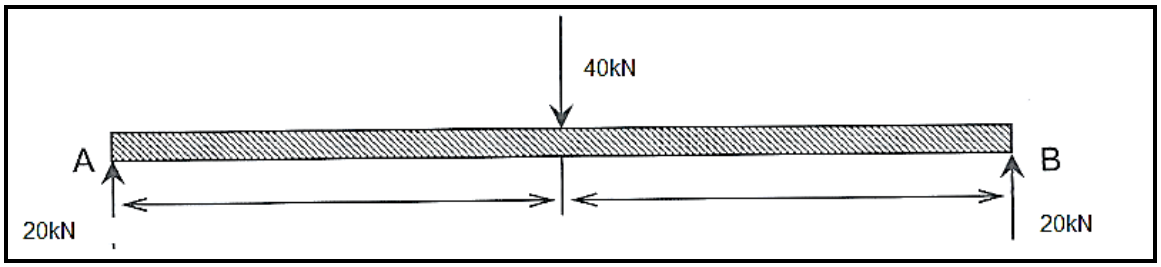


FIGURE 3 (2)

[20]

QUESTION 2: REINFORCED-CONCRETE DETAILING AND SPECIFICATIONS

2.1 State the purpose of drawings in reinforced-concrete detailing. (4)

2.2 Name THREE structural members used in hydrology. (3)

2.3 Briefly explain the position of typical culvert reinforcement. (2)

2.4 Explain how pile caps are similar to beams. (2)

2.5 Explain the concept of symmetrical pile caps. (2)

2.6 What is the maximum number of piles a single pile cap can accommodate? (1)

2.7 Outline the main components of a cantilever retaining wall. (3)

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2.8 Name the structural members in FIGURE 4.

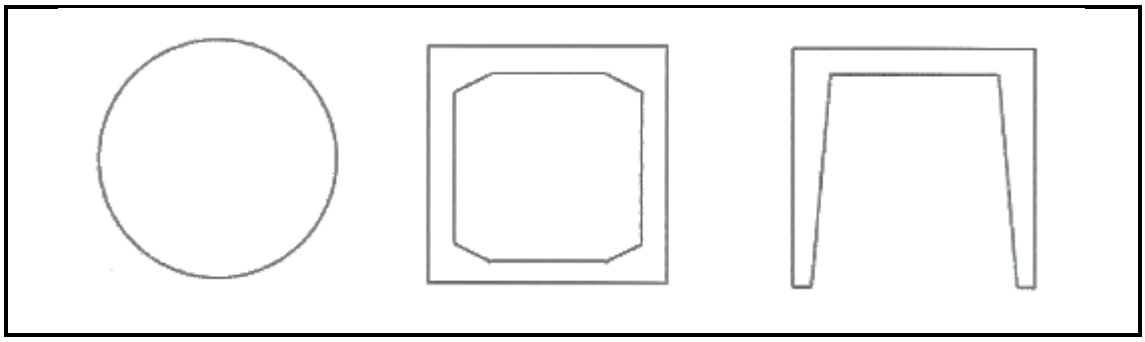


FIGURE 4 (3)
[20]

QUESTION 3: STRUCTURAL STEEL WORK DETAILING AND SPECIFICATIONS

3.1 Calculate the member forces acting on joint A in the truss in FIGURE 5.

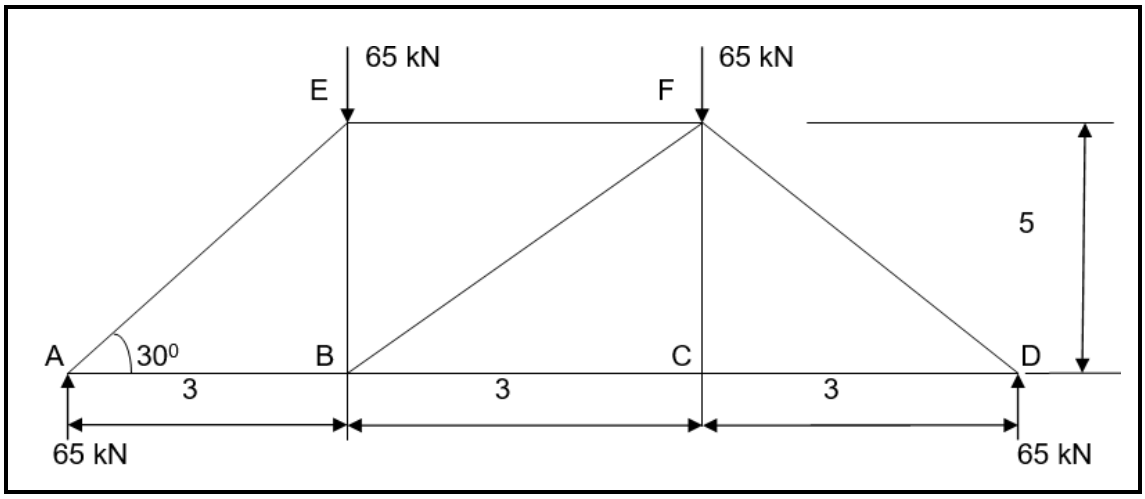


FIGURE 5 (6)

3.2 Use the calculation in QUESTION 3.1 to determine which members are in tension and which are in compression. (2)

3.3 Given that the maximum bending moment is 405 kN and the maximum shear force is 90 kN, study the truss details in FIGURE 6 and calculate:

3.3.1 The maximum compressive force on the top chord (2)

3.3.2 The maximum tensile force of diagonal member EC (3)

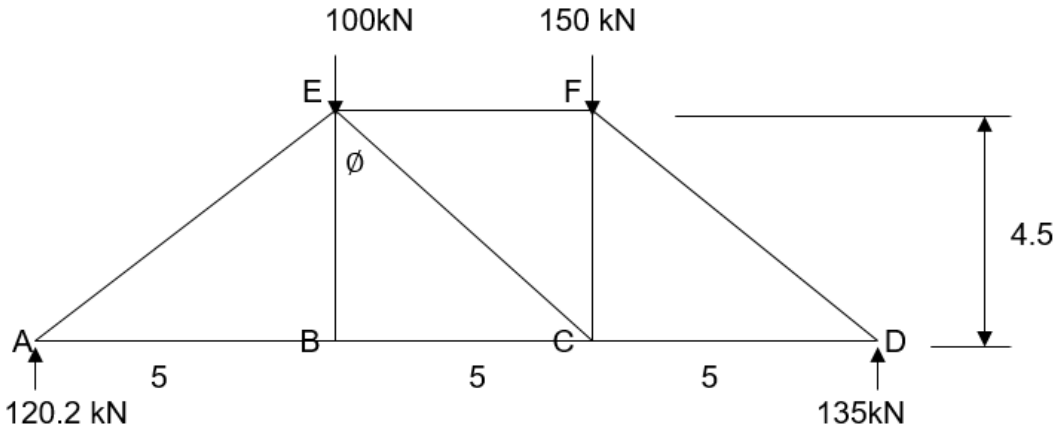


FIGURE 6

3.4 Explain the TWO most common methods of connecting steel sections. (4)

3.5 List THREE methods of detailing steps in a structural steel work stairway. (3)
[20]

QUESTION 4: PROPERTIES OF STRUCTURAL STEEL WORK AND REINFORCED CONCRETE

4.1 List THREE factors that influence the stiffening of concrete. (3)

4.2 Explain why seawater should never be used in concrete. (1)

4.3 List the THREE pieces of equipment used to perform a slump test. (3)

4.4 What would go wrong if a bending schedule is not used? (5)

4.5 Compare *thermal cracks* and *plastic-shrinkage concrete cracks*. (3)
[15]

TOTAL: 75

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ADDENDUM

TABLE 1.1 PROPERTIES OF H-SECTIONS

H section	m (kg/m)	h (mm)	b (mm)	t_w (mm)	t_f (mm)	h_w (mm)	A (10^3 mm^2)	Z_e (10^3 mm^2)	Z_{pl} (10^3 mm)	r (mm)
254 × 254 × 73	73,1	254,2	254,0	8,6	14,2	200	9,29	896	990	64,6
254 × 254 × 89	88,9	250,4	255,9	10,5	17,3	200	11,4	1 100	1 230	65,2
254 × 254 × 107	107	266,7	258,3	13	20,5	200	13,7	1 310	1 480	65,7
254 × 254 × 132	132	276,4	261,0	15,6	25,1	201	16,8	1 620	1 860	66,6
254 × 254 × 167	167	289,1	264,5	19,2	31,7	200	212,2	2 070	2 420	67,9
305 × 305 × 97	96,9	307,8	304,8	9,9	15,4	247	12,3	1 440	1 590	76,8
305 × 305 × 118	118	314,5	306,8	11,9	18,7	247	15,0	1 760	1 950	77,6
305 × 305 × 137	137	320,5	308,7	13,8	21,7	247	17,4	2 040	2 290	78,2
305 × 305 × 158	158	327,2	310,6	15,7	25	247	20,1	2 360	2 670	78,9

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FORMULA SHEET

$F = ma$

$W = mg$

$M_B = F_A d_{AB}$

$\sum Fy = 0$

$\sum Fx = 0$

$\sum M = 0$

$\sigma = \frac{F}{A}$

$F_C = F_S$

$M = F_S \times$ perpendicular distance

$M = F_C \times$ perpendicular distance

$ax^2 + bx + c = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$UDL = Y_{CON} \times$ cross-sectional area

$F_C = 0{,}45\sigma cb(0{,}9x)$

$F_C = 0{,}45\sigma bh$

$F_S = 0{,}87\sigma y A_{st}$

$F_S = 0{,}73\sigma y A_{sc}$

$N = F_C + F_S$

$Q_B A - N - Y_S(1)(A) - Y_C(h)(A) = 0$

$A = \frac{N}{QB - YS(1) - YC(h)}$

$Q_U = \frac{UDL}{A}$

$M_r = \emptyset Z_{Pl} f_y$

$M_r = \emptyset Z_e f_y$

Self-weight of a beam $= m \times a$

$\lambda = \frac{KL}{r}\sqrt{\frac{fy}{\pi^2 E}}$

$C_r = \emptyset A f_y (1 + \lambda^{2,68})^{-1/1,34}$

$T_r = \emptyset A fy$

$F_{AE}\cos\theta + F_{AC} = 0$

$F_{AE} = -\frac{100}{\sin 30}$

$F_{AB}\sin\theta - F_{BE} = 0$

Maximum compressive force $= M_{MAX}/h$

Maximum tensile force $= F_{SMAX}/\cos\emptyset$

$F_R = F_1 + F_2 + F_3$

$0{,}405\sigma_C bx = 0{,}87\sigma_y A_{st}$

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